When he asks “what features?”

Say:

“I’m using **classical computer vision**, not ML training. By features, I mean **geometric and pixel-based properties**, such as continuous straight lines for axes, connected pixel clusters for symbols, and centroids of those clusters for data points.”

When he asks “how does it work?”

Say:

“The pipeline is rule-based: grayscale → thresholding → line detection for axes → contour detection for symbols → centroid calculation → pixel-to-value mapping.”

When he asks about accuracy:

Say:

“Accuracy is bounded by pixel resolution. I’ll estimate numerical error by mapping pixel distance to axis scale and validating against hand-extracted control data.”

“I noticed an existing exploratory notebook and am using it only as contextual reference.”

STEP 1 (Hours 1–2): Ignore junk & open real folder

What to DO

1. Open Initial_set.zip
2. IGNORE:
    - __MACOSX
    - .DS_Store
    - files starting with ._
3. Open **only** the Initial_set folder

What to THINK

"Each folder is one paper. I only need one image."

What to SAY (if asked)

"I filtered out system files and focused on the actual paper folders."

⏱️ **Time to log:** 2 hours

STEP 2 (Hours 3–4): Pick ONE paper folder

What to DO

1. Open folders **one by one**
2. Look only at images
3. Pick a folder that has:
    - One simple plot
    - Clear dots
    - Clear axes

⚠️ Don't overthink. First decent one wins.

What to SAY

"I selected a representative plot with clear axes and data points to develop the initial pipeline."

⏱️ **Time to log:** 2 hours
🟪 Total so far: **4 hours**

STEP 3 (Hours 5–7): Read plots, NOT chemistry

What to DO

Open the Bailey & Winey review paper.

ONLY:

- Look at **figures**
- Ignore equations
- Ignore theory
- Ask:
    - Is axis linear or log?
    - Are there multiple curves?
    - Are points circular?

What to SAY

“I reviewed representative figures to understand plot structure rather than material theory.”

Observed plots:

- Mostly log–log axes

- Data shown as discrete points

- Axes are straight and continuous

- Legends sometimes overlap plot area